

Modelling Toeplitz Networks with SageMath

A Toeplitz network refers to a graph that has a comparable regularity in its structure. SageMath is an excellent tool for facilitating the creation, analysis, and visualisation of graphs. Hence, SageMath can be used to effectively model Toeplitz networks and get insights into their structural characteristics, leading to advancements in network design and analysis.



Toeplitz networks are a category of graphs characterised by an adjacency matrix with a consistent diagonal arrangement. In essence, each row of the matrix is a shifted variant of the preceding row. These networks are commonly utilised in various fields, including signal processing and communication systems, where the interconnections among components display a systematic and repetitive structure.

In graph theory, domination refers to a set of vertices in a graph where every other vertex is either part of this set or is connected to at least one vertex within it. This concept is important because it helps identify key nodes that influence the network, making it a useful tool for optimisation problems, network design, and understanding the structural dynamics of systems.

Analysing domination in Toeplitz networks is essential because of their widespread application in modelling systems with repetitive patterns. Understanding the dominating sets in these networks can improve signal transmission efficiency, refine communication protocols, and help create robust systems. Additionally, this research area is important in

various domains, including social network analysis, power grid optimisation, and sensor networks.

To effectively model and analyse these concepts, SageMath serves as an exceptional tool. It offers a comprehensive environment for engaging with graph theory, featuring functions specifically designed for modelling Toeplitz networks and algorithms for investigating domination. With SageMath, researchers and practitioners can simulate, analyse, and visualise intricate network behaviours, making it an invaluable resource for exploring these subjects.

Understanding the Toeplitz network

A Toeplitz network draws its inspiration from the mathematical idea of a Toeplitz matrix, characterised by having constant values along each descending diagonal from left to right. In graph theory, a Toeplitz network describes a graph that displays a comparable regularity in its structure, where the connections (edges) adhere to a repeating or periodic arrangement across its rows or columns.

A Toeplitz matrix is a special type of matrix in which all the elements along any diagonal, extending from the